

File permissions in Linux

Project description

In this project, I examined and managed file and directory permissions in a Linux environment for a large organization's research team. Using the Bash shell, I reviewed existing permissions to ensure they matched the appropriate authorization levels and modified them where necessary to remove unauthorized access and strengthen system security.

Check file and directory details

```
researcher2@b98f5bf8e016:~/projects$ cd /home/researcher2/projects
researcher2@b98f5bf8e016:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  9 23:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 10 00:59 ..
-rw--w---- 1 researcher2 research_team  46 Jun  9 23:44 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  9 23:44 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun  9 23:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_t.txt
researcher2@b98f5bf8e016:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  9 23:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 10 00:59 ..
-rw--w---- 1 researcher2 research_team  46 Jun  9 23:44 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  9 23:44 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun  9 23:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_t.txt
researcher2@b98f5bf8e016:~/projects$
```

The `ls -la` command lists all files and directories including hidden files, along with their permissions, ownership, and other details.

Describe the permissions string

The 10-character string `-rw-rw-r--` represents file permissions. The first character `-` indicates it is a regular file. The next three characters `rw-` show the user has read and write permissions. The following three `rw-` show the group has read and write permissions. The final three `r--` show others only have read permission.

Change file permissions

```
researcher2@b98f5bf8e016:~/projects$ chmod g-r project_m.txt
researcher2@b98f5bf8e016:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  9 23:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 10 00:59 ..
-rw--w---- 1 researcher2 research_team  46 Jun  9 23:44 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  9 23:44 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun  9 23:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_t.txt
researcher2@b98f5bf8e016:~/projects$

researcher2@b98f5bf8e016:~/projects$ chmod o-w project_k.txt
researcher2@b98f5bf8e016:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  9 23:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 10 00:59 ..
-rw--w---- 1 researcher2 research_team  46 Jun  9 23:44 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  9 23:44 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun  9 23:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_t.txt
researcher2@b98f5bf8e016:~/projects$
```

The `chmod o-w` command removes write permissions from the owner type "other". The `chmod g-r` command removes read permissions from the group.

Change file permissions on a hidden file

```
researcher2@b98f5bf8e016:~/projects$ chmod u-w,g-w .project_x.txt
researcher2@b98f5bf8e016:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  9 23:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 10 00:59 ..
-r----- 1 researcher2 research_team  46 Jun  9 23:44 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  9 23:44 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun  9 23:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_t.txt
researcher2@b98f5bf8e016:~/projects$
```

The `chmod u-w,g-w` command removes write permissions from both the user and the group on the hidden file `.project_x.txt`.

Change directory permissions

```
researcher2@b98f5bf8e016:~/projects$ chmod g-x drafts
researcher2@b98f5bf8e016:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  9 23:44 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 10 00:59 ..
-r----- 1 researcher2 research_team  46 Jun  9 23:44 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jun  9 23:44 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun  9 23:44 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  9 23:44 project_t.txt
researcher2@b98f5bf8e016:~/projects$
```

The `chmod g-x` command removes execute permissions from the group on the `drafts` directory, restricting access to only the `researcher2` user.

Summary

I used the `ls -la` command to check all file and directory permissions, including hidden files, in the `projects` directory. I then used the `chmod` command to remove unauthorized write permissions from `project_k.txt`, restrict group access on `project_m.txt`, remove write permissions from the hidden file `.project_x.txt`, and remove group execute permissions from the `drafts` directory. These changes ensured that only authorized users have appropriate access to sensitive files and directories.